Sampling methods

Key Terms

- In statistics, we generally want to study a population. You can think of a population as a collection of persons, things, or objects under study.

  To study the population, we select a sample. The idea of sampling is to select a portion (or subset) of the larger population and study that portion (the sample) to gain information about the population.

Data are the result of sampling from a population

- Because it takes a lot of **time and money** to examine an entire population, sampling is a very practical technique.
- If you wished to compute the overall grade point average at your school, it would make sense to select a sample of students who attend the school. The **data** collected from the sample would be the students' grade point averages. In presidential elections, opinion poll samples of 1,000–2,000 people are taken. The opinion poll is supposed to *represent the views of the people in the entire country*.

Key Terms

- From the sample data, we can calculate a statistic. A **statistic** is a number that represents a property of the sample.

  For example, if we consider one math class to be a sample of the population of all math classes, then the average number of points earned by students **in that one math class** at the end of the term is an example of a statistic.

  The statistic is an estimate of a population parameter. A **parameter** is a numerical characteristic of the **whole population** that can be estimated by a statistic. Since we considered all math classes to be the population, then the average number of points earned per student over all the math classes is an example of a parameter.

representative sample: The statistics accurately estimates parameters.

Key Terms

- A **variable**, usually notated by capital letters such as X and Y, is a characteristic or measurement that can be determined for each member of a population.
- Variables may be **numerical** or **categorical**.
  - **Numerical variables** take on values with equal units such as weight in pounds and time in hours.
  - **Categorical variables** place the person or thing into a category.

- **Data** are the actual values of the variable. They may be numbers or they may be words. **Datum** is a single value.

Example

Determine what the key terms refer to in the following study. We want to know the average (mean) amount of money first year college students spend at ABC College on school supplies that do not include books. We randomly surveyed 100 first year students at the college. Three of those students spent $150, $200, and $225, respectively.

- The **population** is all first year students attending ABC College this term.
- The **sample** could be all students enrolled in one section of a beginning statistics course at ABC College (although this sample may not represent the entire population).
- The **parameter** is the average (mean) amount of money spent (excluding books) by first year college students at ABC College this term.
- The **statistic** is the average (mean) amount of money spent (excluding books) by first year college students in the sample.
- The **variable** could be the amount of money spent (excluding books) by one first year student. Let X = the amount of money spent (excluding books) by one first year student attending ABC College.
- The **data** are the dollar amounts spent by the first year students. Examples of the data are $150, $200, and $225.

Sampling

- Gathering information about an entire population often costs too much or is virtually impossible. Instead, we use a sample of the population. **A sample should have the same characteristics as the population it is representing.**
- There are several different methods of **random sampling**. In each form of random sampling, each member of a population initially has an equal chance of being selected for the sample.
  - **Simple Random sampling**
  - **Stratified Sampling**
  - **Cluster Sampling**
  - **Systematic Sampling**
- The easiest method to describe is called a **simple random sample**. Any group of n individuals is equally likely to be chosen as any other group of n individuals if the simple random sampling technique is used. In other words, each sample of the same size has an equal chance of being selected.
- For example, suppose Lisa wants to form a four person study group (herself and three other people) from her pre-calculus class, which has 31 members not including Lisa. To choose a simple random sample of size three from the other members of her class, Lisa could put all 31 names in a hat, shake the hat, close her eyes, and pick out three names.

Stratified Sampling

- To choose a stratified sample, **divide the population into groups called strata** and then take a proportionate number from each stratum. For example, you could stratify (group) your college population by department and then choose a proportionate simple random sample from each stratum (each department) to get a stratified random sample.
- To choose a simple random sample from each department, number each member of the first department, number each member of the second department, and do the same for the remaining departments. Then use simple **random sampling** to choose proportionate numbers from the first department and do the same for each of the remaining departments.
- Those numbers picked from the first department, picked from the second department, and so on represent the members who make up the stratified sample.

Cluster Sampling

- To choose a **cluster sample**, divide the population into clusters (groups) and then **randomly select some of the clusters**.

  All the members from these clusters are in the cluster sample. For example, if you randomly sample four departments from your college population, the four departments make up the cluster sample. Divide your college faculty by department. The departments are the clusters. Number each department, and then choose four different numbers using simple random sampling. All members of the four departments with those numbers are the cluster sample.

Systematic Sampling

- To choose a **systematic sample**, randomly select a starting point and take every n_{th} piece of data from a listing of the population. For example, suppose you have to do a phone survey. Your phone book contains 20,000 residence listings. You must choose 400 names for the sample.
- Number the population 1–20,000 and then use a simple random sample to pick a number that represents the first name in the sample. Then choose every fiftieth name thereafter until you have a total of 400 names (you might have to go back to the beginning of your phone list). Systematic sampling is frequently chosen because it is a simple method.

Sampling

- Sampling with/without Replacement
  - Sampling without replacement instead of sampling with replacement becomes a mathematical issue only when the **population is small**.

- Biased Sampling
  - A **sampling bias** is created when a sample is collected from a population and some members of the population are not as likely to be chosen as others (remember, each member of the population should have an equally likely chance of being chosen). When a sampling bias happens, there can be **incorrect conclusions** drawn about the population that is being studied.